

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that:
Packet was undeliverable to port 53 of the DNS server.

This is based on the results of the network analysis, which show that the ICMP echo reply returned the error message:

Unreachable length The port noted in the error message is used for: The word "unreachable" in the message indicates the UDP message requesting an IP address for the domain "www.yummyrecipesforme.com" did not go through to the DNS server because no service was listening on the receiving DNS

The port noted in the error message is used for:
53

The most likely issue is:
No service was listening on the receiving DNS port.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

Time incident occurred:
First incident 13:24:36

Explain how the IT team became aware of the incident:
Attempted to visit the website and you also receive the error "destination port unreachable."

Explain the actions taken by the IT department to investigate the incident:
To troubleshoot the issue, you load your network analyzer tool, tcpdump, and attempt to load the webpage again. To load the webpage, your browser sends a query to a DNS server via the UDP protocol to retrieve the IP address for the website's domain

name; this is part of the DNS protocol. Your browser then uses this IP address as the destination IP for sending an HTTPS request to the web server to display the webpage. The analyzer shows that when you send UDP packets to the DNS server, you receive ICMP packets containing the error message: "udp port 53 unreachable."

Note key findings of the IT department's investigation (i.e., details related to the port affected, DNS server, etc.):

13:24:32.192571 IP 192.51.100.15.52444 > 203.0.113.2.domain: 35084+A?
yummyrecipesforme.com. (24)

13:24:36.098564 IP 203.0.113.2 > 192.51.100.15: ICMP 203.0.113.2 udp port 53
unreachable length 254

Note a likely cause of the incident:

no service was listening on the receiving DNS port